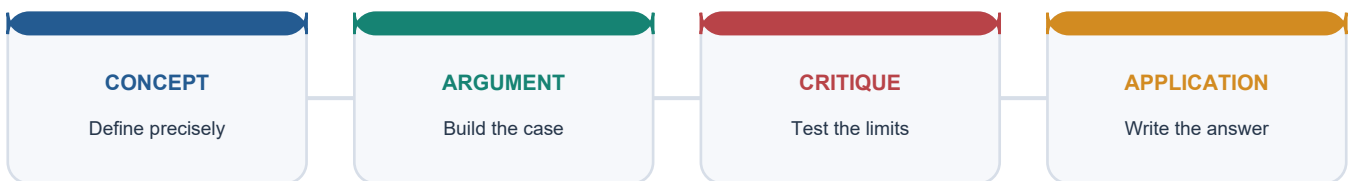


COMPLETE LEARNING SESSION

Arithmetic-and-Commercial-Math - CSAT Learner-v2 Semantic Successor

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing

LEARNING PATH



Deterministic fallback visual: move from precise concepts through argument and critique to exam application.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

CONTENTS / SESSION INDEX

Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

Arithmetic-and-Commercial-Math - Complete CSAT Learning Session 5

BASIC LEARNING SESSION 5

Twelve-panel ASCII master flow	5
Canonical Basic owner	7
Arithmetic and Commercial Math - MUST-DO	7
1. Visual foundation	7
2. Essential formulas	7
3. Method	8
4. Conditions to respect	8
5. Core commercial workflows	8
6. Original solved examples	9
7. Must-Know facts	9
8. Common traps	9
9. Quick checks	10
10. Mini-drill (with answers and explanations)	10
11. Timed mixed practice, diagnosis and retry gate	10
12. Study links	11
2026 PYQ Integration	11
Recent PYQ Integration (2024-2025)	13
Historical PYQ Integration (2018-2023)	15
Semantic-completeness closure - 2026-09-06	19

BASIC MCQS / REMEDIATION 20

Diagnostic and core set	20
Q1. Practice variant 1: What is 15% of 200?	20
Q2. Practice variant 1: A value of 1000 rises by 10% and then falls by 5%. What is the final value?	20
Q3. Practice variant 1: Groups of 20 and 30 have averages 40 and 60. What is their combined average, rounded to the nearest integer?	20
Q4. Practice variant 1: An article costs Rs 500 and is sold at 20% profit. What is the selling price?	20
Q5. Practice variant 1: Find the simple interest on Rs 1000 at 5% per annum for 2 years.	20

Q6. Practice variant 1: Find the compound interest on Rs 1000 at 10% per annum for 2 years, compounded annually.	20
Q7. Practice variant 1: Rs 360 is divided in the ratio 2:3. What is the first share?	20
Q8. Practice variant 1: A invests Rs 1000 for 12 months and B invests Rs 1500 for 8 months. Out of profit Rs 1000, what is A's share, rounded to the nearest rupee?	20
Q9. Practice variant 2: What is 20% of 220?	20
Q10. Practice variant 2: A value of 1000 rises by 11% and then falls by 6%. What is the final value?	21
Q11. Practice variant 2: Groups of 21 and 31 have averages 41 and 61. What is their combined average, rounded to the nearest integer?	21
Q12. Practice variant 2: An article costs Rs 550 and is sold at 20% profit. What is the selling price?	21
Q13. Practice variant 2: Find the simple interest on Rs 1200 at 6% per annum for 2 years.	21
Q14. Practice variant 2: Find the compound interest on Rs 1000 at 15% per annum for 2 years, compounded annually.	21
Q15. Practice variant 2: Rs 420 is divided in the ratio 3:3. What is the first share?	21
Q16. Practice variant 2: A invests Rs 1000 for 12 months and B invests Rs 1500 for 9 months. Out of profit Rs 1000, what is A's share, rounded to the nearest rupee?	21
Remediation protocol	21
PYQS AND ANSWER PRACTICE	21
Verified 2024-2026 Set-A demand and key ledger	22
Timed mixed transfer	22
Q33. Practice variant 5: What is 20% of 280?	22
Q34. Practice variant 5: A value of 1000 rises by 14% and then falls by 9%. What is the final value?	22
Q35. Practice variant 5: Groups of 24 and 34 have averages 44 and 64. What is their combined average, rounded to the nearest integer?	23
Q36. Practice variant 5: An article costs Rs 700 and is sold at 20% profit. What is the selling price?	23
Q37. Practice variant 5: Find the simple interest on Rs 1800 at 9% per annum for 2 years.	23
Q38. Practice variant 5: Find the compound interest on Rs 1000 at 10% per annum for 2 years, compounded annually.	23
Q39. Practice variant 5: Rs 600 is divided in the ratio 3:3. What is the first share?	23
Q40. Practice variant 5: A invests Rs 1000 for 12 months and B invests Rs 1500 for 9 months. Out of profit Rs 1000, what is A's share, rounded to the nearest rupee?	23
OPTIONAL ADVANCED DEPTH - NOT REQUIRED FOR A CORE ANSWER	23
Arithmetic and Commercial Math - ADVANCED	23
1. Architecture	23
2. Advanced tools with conditions	24
3. Harder methods (worked)	24
4. Time-saving techniques (safe conditions)	25
5. Boundary cases	25

6. Advanced traps	25
7. Error analysis	25
8. Advanced drill (with full solutions)	26
9. PYQ-pattern notes (2024-2026, Set A)	26
10. Study links	26
Historical PYQ Integration (2018-2023)	27
CONSOLIDATED REGISTER NOTES	31
Complete revision spine	31
Ownership and close-option firewall	31
Timed answer route	31

Arithmetic-and-Commercial-Math - Complete CSAT Learning Session

Identity: csat-03: learner-v2:g1 **Generation date:** 2026-09-06 **Approval:** false **Official syllabus anchor:** Basic numeracy at Class X level.

Source	SHA-256 at generation
upsc-ai-kit\knowledge\CSAT\basic\03_Arithmetic-and-Commercial-Math.md	9283043c44d58ac8207c7b9ab39d5729bb80481950297680a22fd88c292a5d54
upsc-ai-kit\knowledge\CSAT\advanced\03_Arithmetic-and-Commercial-Math.md	0bdebe411b5f61ec646b126510ffe7bfb9c9fcf3f6dd322902b2b9e3bfe99cc8
upsc-ai-kit\knowledge\CSAT\00_Master-Framework.md	aaa2016904629d268db79a5798150ef8b9a1386fd1edb2ec737062f2cc9e3fcf
upsc-ai-kit\knowledge\CSAT\OFFICIAL-UPSC-SYLLABUS-MAPPING.md	80dd81eb9ccd7e977570eb86cac69418178dc665d981094c6438a95ca4b8b8a8
upsc-ai-kit\knowledge\CSAT\00_Question-Audit-Ledger.md	fbecba6a750ae88aba541dd1ef1379e6e73841ac7a4285b99bec8fd59bea7e6f

The canonical Basic owner is taught first. Optional Advanced material is isolated after practice. All official-PYQ references preserve the repository's supplied/provisional key labels; no unavailable wording, official explanation or official key is invented.

BASIC LEARNING SESSION

Twelve-panel ASCII master flow

PANEL 01 - RATIO LANGUAGE	
Convert comparisons into common units and scalable parts.	
PANEL 02 - PROPORTION AND VARIATION	
Distinguish direct from inverse change.	
PANEL 03 - PERCENT FOUNDATION	
Percent means per hundred; identify the base before operating.	
PANEL 04 - SUCCESSIVE CHANGE	
Multiply factors; do not add percentages blindly.	
PANEL 05 - AVERAGE	
Use total divided by count and preserve group weights.	
PANEL 06 - PROFIT, LOSS AND DISCOUNT	
Keep CP, SP and MP bases distinct.	
PANEL 07 - INTEREST	
Derive SI from principal-rate-time and CI from repeated growth.	
PANEL 08 - MIXTURES	
Use conservation of quantity or alligation with validity checks.	
PANEL 09 - PARTNERSHIP	
Profit share follows capital multiplied by time.	
PANEL 10 - AGES	
Translate one timeline consistently.	
PANEL 11 - ESTIMATION AND OPTIONS	
Use smart numbers only when ratios remain invariant.	
PANEL 12 - VERIFICATION	
Reverse the change, recompute totals and inspect the percentage base.	

Arithmetic and Commercial Math - MUST-DO

Subject: CSAT | **Tier:** Must-Do (foundation) | **Family:** Basic numeracy. **Core skill:** ratio, percentage, average, mixture, profit-loss, interest, partnership and ages - solved by **scaling and multipliers**, not long working. **Grounded in:** audited UPSC CSAT PYQ sets 2024-2026 (Set A). Drills are **computationally verified**. **FACT:** = rule/fact | **ANALYSIS:** = guidance | $\square$ = original example. *Companion: Advanced Drill.*

1. Visual foundation

WORD PROBLEM
|
translate to ONE relation:
|
part/whole (ratio) | per-hundred (percentage) | total/count (average)
|
apply as a MULTIPLIER (x1.2 for +20%, x0.8 for -20%)
|
combine multipliers for successive changes -> read the answer

Core proposition: ratios and percentages are **multipliers**; chaining multipliers replaces most step-by-step arithmetic.

2. Essential formulas

Concept	Exam-ready formula
FACT: Percentage change	For a change of p%, $\text{New} = \text{Old} \times (1 \pm p/100)$. Successive changes multiply, they do not add.
FACT: Percentage point	A change between two percentages : $50\% - 40\% = 10$ percentage points, which is a 25% relative rise. Never write "10%" for it.
FACT: Ratio split	Divide total by the sum of ratio parts to get one "part".
FACT: Average	$\text{Average} = \text{Sum} / \text{Count}$; $\text{Sum} = \text{Average} \times \text{Count}$.
FACT: Weighted average	$(w_1 a_1 + w_2 a_2 + \dots) / (w_1 + w_2 + \dots)$. Equals the plain mean only when all weights are equal.
FACT: Alligation	For $a < m < b$, quantity at value a : quantity at value b is $(b - m) : (m - a)$. At an endpoint, one component has zero quantity; outside $[a, b]$, the target is infeasible.
FACT: Profit / loss	$\text{SP} = \text{CP} \times (1 \pm \text{profit\%/loss\%})$; profit% is on CP .
FACT: Simple interest	$\text{SI} = \text{P} \times \text{R} \times \text{T} / 100$ - interest on the original principal every period, so it grows linearly .
FACT: Compound interest	$A = \text{P} \times (1 + \text{R}/100)^T$; $\text{CI} = A - \text{P}$ - interest on the running amount , so it grows geometrically .
FACT: Non-annual compounding	For k compoundings a year use $A = \text{P} \times (1 + \text{R}/(100k))^{(kT)}$ - halve the rate and double the periods for half-yearly, quarter and quadruple for quarterly.
FACT: Partnership	Profit share $\square$ capital $\times$ time.

Routing boundary: conditional probability is owned by Topic 06. This file uses percentages and ratios that may be prerequisites, but does not treat a probability demand as Commercial Math.

MEMORY: Multiplier trap: a +10% then -10% is **not** zero. $1.10 \times 0.90 = 0.99 \rightarrow$ a **1% net fall**.

MEMORY: **Percentage-point trap**: "the pass rate rose from 40% to 50%" is **+10 percentage points** and **+25%** relative - two different true numbers for one change. Read which one the stem wants.

3. Method

1. Name the quantity asked and the relation type (ratio / percent / average / mixture / commercial).
2. Convert every change to a **multiplier** or a **part**.
3. Combine (multiply multipliers; add parts), then scale to the total.
4. Verify by **magnitude** (is a discount answer smaller than the price? is a share $\leq$ total?).

4. Conditions to respect

- ANALYSIS: Profit/loss and markup percentages are on **CP** unless a **markup on MP** or **discount on MP** is stated - read the base carefully.
- ANALYSIS: Successive percentages **compound**; only convert to a single number via the product of multipliers, never by adding the percentages.
- ANALYSIS: **A percentage change always needs its base named**. The same absolute move is a different percentage on a different base, which is why +25% then -10% does not return to the start.
- ANALYSIS: Averages hide distributions - an average is unchanged by swaps that keep the sum.
- ANALYSIS: $A = P (1 + R/100)^T$ assumes **annual** compounding and a **constant** rate; use the k-period form otherwise, and split the calculation if the rate changes between years.
- ANALYSIS: $CI - SI = P (R/100)^2$ holds **only** for exactly **2 years**, annual compounding, same rate.

5. Core commercial workflows

5.1 Direct and inverse variation

If $y \propto x$, then y/x is constant: doubling x doubles y . If $y \propto 1/x$, then xy is constant: doubling x halves y . First name what is held fixed; do not call every changing pair "inverse."

Worked example. 8 workers make 240 units in the same time at the same rate. 12 workers make $240 \times 12/8 = 360$ units (direct variation). If 8 identical workers finish a job in 15 days, 12 workers finish in $8 \times 15/12 = 10$ days (inverse variation, with equal productivity).

5.2 Marked-price and discount workflow

CP $\rightarrow$ (markup, if given) $\rightarrow$ MP $\rightarrow$ (each discount multiplier) $\rightarrow$ SP
profit / loss percentage = $(SP - CP) / CP \times 100$

Worked example. Cost price is 500, markup 20%, then discounts 10% and 5%. $MP = 500 \times 1.20 = 600$; $SP = 600 \times 0.90 \times 0.95 = 513$. Profit = 13, hence profit percentage **2.6%** on CP. Do not subtract discounts from CP or add successive discounts.

5.3 Correcting an average

If one recorded value r should have been c in a group of n , corrected average = old average + $(c - r)/n$. For several errors, add every (correct - recorded) difference to the old sum first.

Worked example. The average of 20 marks is recorded as 48 after 35 was entered instead of 53. Correct average = $48 + (53 - 35)/20 = 48.9$.

5.4 Compound interest with a withdrawal or instalment

For a deposit, move chronologically: compound the current balance for one period, then add or subtract the cash flow at the stated time. The usual $P(1+r)^T$ formula applies only when no intermediate cash flow changes principal.

Worked example. Deposit 10,000 at 10% compounded annually; withdraw 1,100 at the end of year 1. After year 1: $11,000 - 1,100 = 9,900$; after year 2: $9,900 \times 1.10 = 10,890$. A withdrawal at the *beginning* of a year would be subtracted before that year's compounding.

6. Original solved examples

□ Example A (ratio)

Divide 3300 among A, B, C in the ratio 2 : 3 : 6. Parts = 11; one part = $3300/11 = 300$. Shares = **600, 900, 1800**. (Verified.)

□ Example B (successive percentage)

A population rises 10% then falls 10%. Net change? $1.10 \times 0.90 = 0.99 \rightarrow$ **1% decrease**. (Verified.)

□ Example C (average)

The average of 5 numbers is 27. One number, 35, is removed. New average of the 4? Sum = 135; new sum = 100; average = $100/4 = 25$. (Verified.)

□ Example D (alligation)

In what ratio mix rice at 30 and 40 per kg to get 34 per kg? $(40 - 34) : (34 - 30) = 6 : 4 = 3 : 2$. (Verified.)

□ Example E (compound interest)

CI on 10000 at 10% for 2 years. $A = 10000 \times 1.1^2 = 12100$; $CI = 12100 - 10000 = 2100$. (Verified.)

Compare SI on the same terms: $10000 \times 10 \times 2 / 100 = 2000$. The 100 difference is exactly $P (R/100)^2 = 10000 \times 0.01$. (Verified.)

□ Example F (weighted average vs plain average)

A class has 30 boys averaging 62 marks and 20 girls averaging 72. Class average = $(30 \times 62 + 20 \times 72) / 50 = (1860 + 1440) / 50 = 3300 / 50 = 66$. The plain mean of 62 and 72 is 67 - wrong, because the groups are not the same size. (Verified.)

□ Example G (percentage point vs percent change)

A district's literacy rate rises from 40% to 50%. The rise is **10 percentage points**. As a **relative** change it is $(50 - 40) / 40 = 25\%$. Both are correct answers to **different** questions - and the options will usually contain both. (Verified.)

7. Must-Know facts

- FACT: $1/2 = 50\%$, $1/3 \approx 33.3\%$, $1/4 = 25\%$, $1/8 = 12.5\%$, $1/20 = 5\%$ - memorise these for speed.
- FACT: Profit% and loss% are always on **cost price**.
- FACT: For 2 years, $CI - SI = P \times (R/100)^2$ (a fast check; 2 years only, annual compounding).
- FACT: **SI is linear in time, CI is geometric** - so CI and SI are equal after **one** period and CI is larger after that (for a positive rate).
- FACT: A **percentage point** is a gap between percentages; a **percent change** is relative to a base.
- FACT: In partnership, equal-time investors share in the **capital ratio**; equal-capital investors share in the **time ratio**.

8. Common traps

- WRONG: Adding successive percentages ($10\% + 10\% = 20\%$). $\rightarrow$ Multiply multipliers ($1.1 \times 1.1 = 1.21$).
- WRONG: Taking profit% on selling price. $\rightarrow$ It is on **cost price**.
- WRONG: Assuming +x% then -x% returns to the start. $\rightarrow$ It gives a net $x^2/100$ % fall.
- WRONG: Saying "the rate rose 10%" when it rose from 40% to 50%. $\rightarrow$ That is **10 percentage points** (and 25% relatively).
- WRONG: Averaging two group averages by a plain mean when the **group sizes differ**. $\rightarrow$ Weight by size.
- WRONG: Averaging two speeds/prices by a plain mean when the **weights differ**. $\rightarrow$ Weight by quantity.
- WRONG: Reading a discount off cost price. $\rightarrow$ Discount is off **marked price**.
- WRONG: Using the annual CI formula when compounding is half-yearly/quarterly. $\rightarrow$ Adjust R and T.

9. Quick checks

- FACT: Can you turn "increases by 25%" into "x1.25" instantly?
- FACT: Given a ratio and a total, can you get one part in one division?
- FACT: Can you write alligation ratio without a diagram?
- FACT: Can you say, for one stated change, both its percentage-point and its percent value?

10. Mini-drill (with answers and explanations)

1. If $A : B = 2 : 3$ and $B : C = 4 : 5$, find $A : B : C$.
2. Two successive discounts of 20% and 10% equal what single discount?
3. The average age of 30 students is 12. Including the teacher, the average of 31 becomes 13. Find the teacher's age.
4. An article sold for 480 at a 20% loss. Find the cost price.
5. A invests 12000 for 6 months and B invests 8000 for 12 months; total profit is 4200. Find B's share.
6. A scheme's approval rating moves from 25% to 30%. State the change (i) in percentage points and (ii) as a percent change.
7. A shop sells 3 kg of rice at 40 per kg and 2 kg at 50 per kg. Find the average price per kg.
8. Find CI - SI on 8000 at 5% per annum for 2 years.

Answers.

1. **8 : 12 : 15.** Make B common (12): $A : B = 8 : 12$, $B : C = 12 : 15$.
2. **28%.** $1 - (0.8 \times 0.9) = 0.28$. (Verified.)
3. **43.** Students' sum = 360; with teacher = $31 \times 13 = 403$; teacher = 43. (Verified.)
4. **600.** $CP = 480 / 0.8 = 600$. (Verified.)
5. **2400.** Capital x time = $72000 : 96000 = 3 : 4$; $B = 4200 \times 4/7 = 2400$. (Verified.)
6. **(i) 5 percentage points. (ii) 20%.** $30 - 25 = 5$ points; $(30 - 25)/25 = 0.20$. Both are correct - for different questions. (Verified.)
7. **44.** Weighted: $(3 \times 40 + 2 \times 50)/5 = 220/5 = 44$. The plain mean of 40 and 50 is **45**, which is wrong because the quantities differ. (Verified.)
8. **20.** $CI = 8000 \times 1.05^2 - 8000 = 820$; $SI = 8000 \times 5 \times 2/100 = 800$; difference = 20, which equals $P (R/100)^2 = 8000 \times 0.0025$. (Verified.)

11. Timed mixed practice, diagnosis and retry gate

Set rule: solve in **12 minutes** without a calculator. For every answer, label the governing relationship first: P percentage multiplier, V variation, W weighted average, M marked-price chain, C cash-flow compounding, or R ratio.

1. Price rises by 40%. By what percentage must consumption fall to keep expenditure unchanged?
2. If 15 machines make 900 parts in 6 hours, how many parts do 10 identical machines make in 9 hours?
3. A 25-member class has recorded average 60. One score 42 was actually 57. Find the corrected average.
4. CP is 800; markup is 25%, followed by discounts 10% and 20%. Find profit/loss percentage.
5. A deposit of 20,000 earns 5% annual CI. At the end of year 1, 1,000 is withdrawn. Find the balance at the end of year 2.
6. Mix 20 kg at 30/kg and 30 kg at 45/kg. Find the mean price/kg.
7. A:B capital ratio is 3:5; A invests 8 months and B 6 months. Divide profit 5,400.
8. A pass rate rises from 48% to 60%. State the percentage-point and relative changes.

Answers. 1. $40/140 = 28\frac{4}{7}\%$ (P); 2. **900** (V, output $\square$ machines x time);
3. $60 + (57 - 42)/25 = 60.6$ (W); 4. $1.25 \times 0.90 \times 0.80 = 0.90$, **10% loss** (M);

5. $(20,000 \times 1.05 - 1,000) \times 1.05 = 21,000$ (C); 6. $(600 + 1,350) / 50 = 39$ (W);
 7. time-weighted ratio $24 : 30 = 4 : 5$, shares **2,400 and 3,000** (R); 8. ****12** percentage points; 25% relative rise** (P).

Miss pattern	Error code	Repair before retry
Added percentages or used the wrong base	C	Draw CP->MP->SP or write every multiplier.
Plain-averaged unequal groups / quantities	W	Rebuild totals as quantity x value.
Reversed direct and inverse variation	V	State the constant ($y/\blacksquare x$ or xy) before numbers.
Compounded a withdrawn/deposited amount for the wrong period	C	Write a year-by-year balance line.
Accurate but too slow	T	Redo using fractions/smart numbers; park at the set ceiling.

Date / set	Score / time	P	V	W	M	C	R	T	Next repair

Gate: score **7/8 within 12 minutes**, then **7/8 on a new mixed set** after any correction before treating Advanced methods as optional enrichment. For full papers, use the Master Framework's qualifying-margin and negative-marking rules; never try to recover a weak Core score with blind commercial-math attempts.

12. Study links

- FACT: Advanced companion - repeated replacement, markup-plus-discount, and equating-ages methods.
- FACT: Number Systems and Number Sense - ratio sense rests on factors.
- FACT: Rates, Motion, Time and Geometry - averages and ratios extend to speed and work.

2026 PYQ Integration

Status: 2026 question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: `_PYQ-ROUTING-CSAT-2026.md`. **Answer-key rule:** The 2026 Prelims and CSAT Set-A keys held locally are **provisional**; no option or answer is recorded or inferred in this integration.

- **Year represented:** 2026
- **Paper(s):** CSAT
- **Routed question demands:** 7

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2026	CSAT	25	Class-average correction	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2026	CSAT	30	Direct/inverse proportion	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	43	Percentage savings change	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	58	Company-worker ratio	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	65	Alloy-mixture ratio	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	78	Partnership capital-duration share	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	79	Repeated-replacement mixture	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.

What this owner must now support

- Class-average correction

- Direct/inverse proportion
- Percentage savings change
- Company-worker ratio
- Alloy-mixture ratio
- Partnership capital-duration share
- Repeated-replacement mixture

This block integrates the 2026 examinable demand and paper metadata. It is kept separate from the 2018-2023 and 2024-2025 blocks and does not convert a provisionally-keyed, answer-free objective question into a solved answer.

Recent PYQ Integration (2024-2025)

Status: 2024-2025 question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: _PYQ-ROUTING-CSAT-2024-2025.md. **Answer-key rule:** The official 2024-2025 Prelims Set-A keys are present in the repository and CSAT Set-A keys are supplied; even so, no option or answer is recorded or inferred in this integration.

- **Years represented:** 2024, 2025
- **Paper(s):** CSAT
- **Routed question demands:** 11

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2024	CSAT	16	Mixture gain	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	25	Partnership profit share	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	26	Successive-percentage comparison	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	27	Inverse-operation error	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	28	Overlapping pass percentages	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2024	CSAT	29	Age ratio	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	36	Proportional weights	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	47	Chained price ratios	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	10	Successive-percentage cycle	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	17	Fuel-cost distance adjustment	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	78	Average-balance relation	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.

What this owner must now support

- Mixture gain
- Partnership profit share
- Successive-percentage comparison
- Inverse-operation error
- Overlapping pass percentages
- Age ratio
- Proportional weights
- Chained price ratios
- Successive-percentage cycle
- Fuel-cost distance adjustment
- Average-balance relation

This block integrates the 2024-2025 examinable demand and paper metadata. It is kept separate from the 2018-2023 block and does not convert an unkeyed/answer-free objective question into a solved answer.

Historical PYQ Integration (2018-2023)

Status: Question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: _PYQ-ROUTING-CSAT-2018-2023.md. **Answer-key rule:** The official 2018-2023 Prelims/CSAT keys are not held locally; no option or answer has been inferred.

- **Years represented:** 2018, 2019, 2020, 2021, 2022, 2023
- **Paper(s):** CSAT
- **Routed question demands:** 40

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2018	CSAT	39	Conditional probability	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	41	Pass-fail percentage marks	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	57	Symmetric profit-loss point	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	68	Capacity proportion share	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	69	Compound interest installment	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	11	Percentage weight reduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	17	Percentage shortfall amount	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	19	Unit price joint purchase	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	34	Percentage marks difference	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	39	Discount from quantity change	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2019	CSAT	40	Mean error correction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	55	Average group weight	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	18	Weighted group average	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	39	Age group average	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	40	Loss-percentage cost recovery	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	47	Successive-discount comparison	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	55	Percentage ratio constraint	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	57	Ratio share difference	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	59	Class average invariance	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	68	Successive mixture replacement	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	69	Innings average difference	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	70	Price-hike quantity reduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	78	Volume-mass unit conversion	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	7	Linear fuel-price crossover	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2021	CSAT	9	Percentage group overlap	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	30	Proportional marks threshold	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	37	Class average transfer effect	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	53	Equal-cost price ratio	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	54	Percentage marks comparison	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	55	Conditional money allocation	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	70	Ratio-share inequality	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	78	Successive percentage change	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	8	Denomination combination validity	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	20	Successive percentage change	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	26	Equal interval spacing	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	45	Word problem total cost	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	64	Election vote percentage	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	69	Mixture proportion comparison	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2022	CSAT	80	Average weight combination	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	44	Compound interest frequency comparison	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

What this owner must now support

- Conditional probability
- Pass-fail percentage marks
- Symmetric profit-loss point
- Capacity proportion share
- Compound interest installment
- Percentage weight reduction
- Percentage shortfall amount
- Unit price joint purchase
- Percentage marks difference
- Discount from quantity change
- Mean error correction
- Average group weight
- Weighted group average
- Age group average
- Loss-percentage cost recovery
- Successive-discount comparison
- Percentage ratio constraint
- Ratio share difference
- Class average invariance
- Successive mixture replacement
- Innings average difference
- Price-hike quantity reduction
- Volume-mass unit conversion
- Linear fuel-price crossover
- Percentage group overlap
- Proportional marks threshold
- Class average transfer effect
- Equal-cost price ratio
- Percentage marks comparison
- Conditional money allocation
- Ratio-share inequality
- Successive percentage change
- Denomination combination validity
- Equal interval spacing
- Word problem total cost

- Election vote percentage
- Mixture proportion comparison
- Average weight combination
- Compound interest frequency comparison

The table integrates the examinable demand and paper metadata. It does not turn an unkeyed objective question into a solved answer, and it does not claim that lexical presence alone proves full conceptual sufficiency.

Semantic-completeness closure - 2026-09-06

Literal syllabus and canonical ownership

- **Literal clause:** Basic numeracy at Class X level.
- **Canonical scope:** Owns ratio, proportion, variation, percentages, averages, mixtures, profit-loss-discount, simple and compound interest, partnership and ages.
- **Cross-topic boundary:** Pure number properties belong to Topic 02; time-work and motion rates to Topic 04; equation sufficiency to Topic 05.

Complete learner route

1. **Ratio language:** Convert comparisons into common units and scalable parts.
2. **Proportion and variation:** Distinguish direct from inverse change.
3. **Percent foundation:** Percent means per hundred; identify the base before operating.
4. **Successive change:** Multiply factors; do not add percentages blindly.
5. **Average:** Use total divided by count and preserve group weights.
6. **Profit, loss and discount:** Keep CP, SP and MP bases distinct.
7. **Interest:** Derive SI from principal-rate-time and CI from repeated growth.
8. **Mixtures:** Use conservation of quantity or alligation with validity checks.
9. **Partnership:** Profit share follows capital multiplied by time.
10. **Ages:** Translate one timeline consistently.
11. **Estimation and options:** Use smart numbers only when ratios remain invariant.
12. **Verification:** Reverse the change, recompute totals and inspect the percentage base.

Verification and hostile-query gate

Every formula is derived from part/whole, multiplier, weighted-total or principal-time logic. Generated answers are recomputed with exact fractions before formatting.

The hostile absence search explicitly tested these families and close-option terms: **ratio; percentage; weighted average; profit; discount; simple interest; compound interest; alligation**. A shortcut is usable only when its stated condition survives; otherwise return to the first-principles representation. Every worked answer must finish with an independent check: passage support, substitution, enumeration, units, bounds, reverse operation or option elimination.

Difficulty and timed progression

1. Foundation: recognise the family and state the governing definition or relation.
2. Core: solve a direct item with a visible representation and one verification.
3. Advanced: combine two mechanisms, test edge cases and reject close distractors.
4. Timed: use classify -> extract -> represent -> execute -> verify -> decide.
5. Remediation: log the error as concept, application, calculation, reading, passage, time or guess; return to the owning subtopic before retesting.

BASIC MCQS / REMEDIATION

Diagnostic and core set

Q1. Practice variant 1: What is 15% of 200?

A. 30 B. 31 C. 29 D. 32

Correct answer: A. Percent means per hundred: base x rate/100.

Q2. Practice variant 1: A value of 1000 rises by 10% and then falls by 5%. What is the final value?

A. 1046 B. 1045 C. 1044 D. 1047

Correct answer: B. Successive changes multiply growth factors; they do not simply cancel.

Q3. Practice variant 1: Groups of 20 and 30 have averages 40 and 60. What is their combined average, rounded to the nearest integer?

A. 53 B. 51 C. 52 D. 54

Correct answer: C. Combine totals, then divide by the combined count.

Q4. Practice variant 1: An article costs Rs 500 and is sold at 20% profit. What is the selling price?

A. 601 B. 599 C. 602 D. 600

Correct answer: D. Profit percent uses cost price as the base.

Q5. Practice variant 1: Find the simple interest on Rs 1000 at 5% per annum for 2 years.

A. 100 B. 101 C. 99 D. 102

Correct answer: A. $SI = \frac{PRT}{100}$, derived by adding the same annual interest on the original principal.

Q6. Practice variant 1: Find the compound interest on Rs 1000 at 10% per annum for 2 years, compounded annually.

A. 211 B. 210 C. 209 D. 212

Correct answer: B. Amount = $P(1+r/100)^n$; subtract principal for compound interest.

Q7. Practice variant 1: Rs 360 is divided in the ratio 2:3. What is the first share?

A. 145 B. 143 C. 144 D. 146

Correct answer: C. One share equals total divided by total ratio-parts.

Q8. Practice variant 1: A invests Rs 1000 for 12 months and B invests Rs 1500 for 8 months. Out of profit Rs 1000, what is A's share, rounded to the nearest rupee?

A. 501 B. 499 C. 502 D. 500

Correct answer: D. Partnership shares are proportional to capital multiplied by time.

Q9. Practice variant 2: What is 20% of 220?

A. 44 B. 45 C. 43 D. 46

Correct answer: A. Percent means per hundred: base x rate/100.

Q10. Practice variant 2: A value of 1000 rises by 11% and then falls by 6%. What is the final value?

A. 1044 B. 1043 C. 1042 D. 1045

Correct answer: B. Successive changes multiply growth factors; they do not simply cancel.

Q11. Practice variant 2: Groups of 21 and 31 have averages 41 and 61. What is their combined average, rounded to the nearest integer?

A. 54 B. 52 C. 53 D. 55

Correct answer: C. Combine totals, then divide by the combined count.

Q12. Practice variant 2: An article costs Rs 550 and is sold at 20% profit. What is the selling price?

A. 661 B. 659 C. 662 D. 660

Correct answer: D. Profit percent uses cost price as the base.

Q13. Practice variant 2: Find the simple interest on Rs 1200 at 6% per annum for 2 years.

A. 144 B. 145 C. 143 D. 146

Correct answer: A. $SI = \frac{PRT}{100}$, derived by adding the same annual interest on the original principal.

Q14. Practice variant 2: Find the compound interest on Rs 1000 at 15% per annum for 2 years, compounded annually.

A. 323 B. 322 C. 321 D. 324

Correct answer: B. $\text{Amount} = P(1 + \frac{r}{100})^n$; subtract principal for compound interest.

Q15. Practice variant 2: Rs 420 is divided in the ratio 3:3. What is the first share?

A. 211 B. 209 C. 210 D. 212

Correct answer: C. One share equals total divided by total ratio-parts.

Q16. Practice variant 2: A invests Rs 1000 for 12 months and B invests Rs 1500 for 9 months. Out of profit Rs 1000, what is A's share, rounded to the nearest rupee?

A. 472 B. 470 C. 473 D. 471

Correct answer: D. Partnership shares are proportional to capital multiplied by time.

Remediation protocol

1. Recompute without options.
2. Name the failed rule or passage phrase.
3. Reject every distractor for a specific reason.
4. Retry with altered numbers, wording or constraints.
5. Advance only after two consecutive correct answers under the time ceiling.

PYQS AND ANSWER PRACTICE

Verified 2024-2026 Set-A demand and key ledger

Year	Q	Neutral verified demand	Key status	Solution architecture
2024	16	Mixture gain	A (supplied)	Conserve the amount of the active component.
2024	25	Partnership profit share	A (supplied)	Compare capital multiplied by time.
2024	26	Successive-percentage comparison	B (supplied)	Fix the base and multiply successive change factors.
2024	27	Inverse-operation error	D (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2024	28	Overlapping pass percentages	B (supplied)	Fix the base and multiply successive change factors.
2024	29	Age ratio	A (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2024	36	Proportional weights	B (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2024	47	Chained price ratios	D (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	10	Successive-percentage cycle	A (supplied)	Fix the base and multiply successive change factors.
2025	17	Fuel-cost distance adjustment	B (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	78	Average-balance relation	A (supplied)	Rebuild the total before and after the change.
2026	25	Class-average correction	B (provisional)	Rebuild the total before and after the change.
2026	30	Direct/inverse proportion	B (provisional)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2026	43	Percentage savings change	B (provisional)	Fix the base and multiply successive change factors.
2026	58	Company-worker ratio	B (provisional)	Add rates and invert the net rate.
2026	65	Alloy-mixture ratio	A (provisional)	Conserve the amount of the active component.
2026	78	Partnership capital-duration share	A (provisional)	Compare capital multiplied by time.
2026	79	Repeated-replacement mixture	B (provisional)	Conserve the amount of the active component.

The table is a non-verbatim routing audit. It records the locally checked Set-A key letter and a valid solving route, but does not pretend that UPSC publishes model solutions. The separate workbook supplies original solved equivalents for every mechanism.

Timed mixed transfer

Q33. Practice variant 5: What is 20% of 280?

A. 56 B. 57 C. 55 D. 58

Correct answer: A. Percent means per hundred: $\text{base} \times \text{rate}/100$.

Q34. Practice variant 5: A value of 1000 rises by 14% and then falls by 9%. What is the final value?

A. 1038 B. 1037 C. 1036 D. 1039

Correct answer: B. Successive changes multiply growth factors; they do not simply cancel.

Q35. Practice variant 5: Groups of 24 and 34 have averages 44 and 64. What is their combined average, rounded to the nearest integer?

A. 57 B. 55 C. 56 D. 58

Correct answer: C. Combine totals, then divide by the combined count.

Q36. Practice variant 5: An article costs Rs 700 and is sold at 20% profit. What is the selling price?

A. 841 B. 839 C. 842 D. 840

Correct answer: D. Profit percent uses cost price as the base.

Q37. Practice variant 5: Find the simple interest on Rs 1800 at 9% per annum for 2 years.

A. 324 B. 325 C. 323 D. 326

Correct answer: A. $SI = PRT/100$, derived by adding the same annual interest on the original principal.

Q38. Practice variant 5: Find the compound interest on Rs 1000 at 10% per annum for 2 years, compounded annually.

A. 211 B. 210 C. 209 D. 212

Correct answer: B. Amount = $P(1+r/100)^n$; subtract principal for compound interest.

Q39. Practice variant 5: Rs 600 is divided in the ratio 3:3. What is the first share?

A. 301 B. 299 C. 300 D. 302

Correct answer: C. One share equals total divided by total ratio-parts.

Q40. Practice variant 5: A invests Rs 1000 for 12 months and B invests Rs 1500 for 9 months. Out of profit Rs 1000, what is A's share, rounded to the nearest rupee?

A. 472 B. 470 C. 473 D. 471

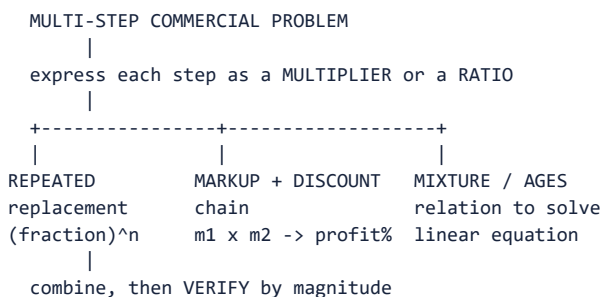
Correct answer: D. Partnership shares are proportional to capital multiplied by time.

OPTIONAL ADVANCED DEPTH - NOT REQUIRED FOR A CORE ANSWER

Arithmetic and Commercial Math - ADVANCED

Subject: CSAT | **Tier:** Advanced Drill | **Family:** Basic numeracy. **Core skill:** multi-step commercial arithmetic - repeated replacement, markup-and-discount chains, equated-ages, expenditure-consumption trade-offs, and two-vessel mixtures. **Grounded in:** audited UPSC CSAT PYQ sets 2024-2026 (Set A). Drills are **computationally verified**. **FACT:** = rule/fact | **ANALYSIS:** = guidance | □ = original example. *Companion: Foundation.*

1. Architecture



Analytical claim: hard arithmetic is a **chain of multipliers plus one linear equation** - never a pile of separate computations.

2. Advanced tools with conditions

Tool	Statement	Condition
FACT: Repeated replacement	After removing and replacing fraction each time, remaining pure part = $\text{initial} \times (1 - \frac{\text{removed}}{\text{total}})^n$.	The vessel is thoroughly mixed, the same volume is removed and replaced each round, and total volume is restored after every round.
FACT: Markup-then-discount	Net multiplier = $(1 + \text{markup}) \times (1 - \text{discount})$; $\text{profit}\% = (\text{net} - 1) \times 100$.	Markup on CP, discount on MP.
FACT: Expenditure = price x quantity	To hold expenditure constant when price rises by r , cut quantity by $r / (1+r)$.	Price and quantity vary inversely.
FACT: Ages	Set present ages as variables; write one equation per time reference.	Keep "years later/ago" consistent on both sides.
FACT: Two-vessel mixture	Combined fraction = weighted average of the two fractions by volume.	Weight by the volume actually taken.

3. Harder methods (worked)

□ Method 1 - repeated replacement

From 40 L of milk, 8 L is removed and replaced with water; repeated 3 times. Milk left? $40 \times (1 - 8/40)^3 = 40 \times (4/5)^3 = 40 \times 0.512 = 20.48 \text{ L. (Verified.)}$

□ Method 2 - markup and discount

Goods marked up 40% and sold at 25% discount. Profit%? $1.40 \times 0.75 = 1.05 \rightarrow 5\% \text{ profit. (Verified.)}$

□ Method 3 - equated ages

A father is thrice his son's age now; in 12 years he will be twice. Let son = s , father = $3s$. $3s + 12 = 2(s + 12) \rightarrow s = 12$, father = 36. **Son 12, father 36. (Verified.)**

□ Method 4 - three different "percentages" from one situation

ANALYSIS: When a **rate** applied to a **changing base** moves, there are **three** true percentage answers. The options will contain more than one of them.

A scheme is approved by 30% of 2000 respondents in year 1 and by 36% of 2500 in year 2.

Question actually asked	Computation	Answer
Change in the approval rate , in percentage points	$36 - 30$	6 percentage points
Percent change in the approval rate	$(36 - 30) / 30$	20%
Percent change in the number of approvers	$600 - 900, (900 - 600) / 600$	50%

(Verified: $0.30 \times 2000 = 600$; $0.36 \times 2500 = 900$.)

- MEMORY: **Escalation rule:** underline the noun the stem attaches "%" to - **rate** or **count** - before you compute anything. Then decide whether it wants a **gap** (percentage points) or a **ratio** (percent change).

4. Time-saving techniques (safe conditions)

- ANALYSIS: **Multiplier chaining** for any sequence of percentage changes. *Safe always.*
- ANALYSIS: **Fraction-power formula** for replacement. *Safe only when the vessel is mixed before every draw, the same quantity is removed and replaced, and the original total volume is restored each round.*
- ANALYSIS: $r/(1+r)$ **shortcut** for the consumption cut that holds expenditure fixed. *Safe when price and quantity are the only variables.*
- ANALYSIS: **Smart-number base (take 100 or the LCM)** for percentage/ratio word problems. *Safe when the answer is a ratio or a percentage, not an absolute value.*

5. Boundary cases

- ANALYSIS: Markup and discount must use the **correct base** (markup on CP, discount on MP); mixing bases is the most common error.
- ANALYSIS: Replacement formula fails if the removed amount changes, the vessel is not mixed before a draw, or the total volume is not restored between rounds.
- ANALYSIS: A weighted average of two mixtures uses the **volumes taken**, not a plain average, unless the volumes are equal.
- ANALYSIS: A percentage applied to a **moving base** yields several different true answers - name the base before computing (Method 4).
- ANALYSIS: Successive discounts multiply; two discounts of d_1 and d_2 are equivalent to a single $d_1 + d_2 - d_1 d_2/100$, never $d_1 + d_2$.

6. Advanced traps

- WRONG: Cutting consumption by $r\%$ (not $r/(1+r)$) to offset an $r\%$ price rise. -> Use $r/(1+r)$.
 - WRONG: Applying discount on cost price. -> Discount is on marked price.
 - WRONG: Adding two successive discounts. -> Multiply the multipliers.
 - WRONG: Reporting a **percent change** when the stem asked for **percentage points**, or the reverse.
- > Underline the noun before computing.
- WRONG: Averaging two mixture ratios directly. -> Weight by volume.
 - WRONG: Averaging two group averages when the groups differ in size. -> Weight by group size.
 - WRONG: In ages, writing "+12" on one side only. -> Add the time to **both** ages.

7. Error analysis

Recurring miss	Master-Framework code	Fix
Wrong base for %/discount	C	Label CP/MP/SP before writing a multiplier.
Adding percentages	A	Always chain multipliers.
Percentage point vs percent	R	Underline "rate" or "count" in the stem first.
Age-equation sign slips	A	Tabulate ages at each time point.
Mixture weighting	X	Use volume x fraction, then divide by total volume.

8. Advanced drill (with full solutions)

ANALYSIS: These items are **new** - they do not repeat Methods 1-4 above or the Foundation file.

1. To keep expenditure unchanged after a 25% price rise, by what % must consumption fall?
2. Vessel 1 has milk : water = 3 : 1; vessel 2 has 5 : 2. Equal volumes are mixed. Find the new milk : water ratio.
3. A trader marks goods 25% above cost and then allows two successive discounts of 10% and 4%. Find the profit percentage.
4. From 40 L of pure milk, 10 L is drawn off and replaced with water; this is done **twice**. How much milk remains?
5. A class of 50 averages 60 marks. The 30 boys average 56. Find the girls' average.
6. An article sold at a 20% loss would have yielded a 10% gain had the selling price been 300 higher. Find the cost price.

Solutions.

1. **20%.** $25/(100 + 25) = 25/125 = 20\%$. (*Verified.*) ANALYSIS: Not 25% - the base changed.
2. **41 : 15.** Milk fraction = $(3/4 + 5/7)/2 = 41/56$; water = $15/56$. (*Verified.*)
3. **8% profit.** Chain the multipliers on **CP**: $1.25 \times 0.90 \times 0.96 = 1.08$. (*Verified.*)
ANALYSIS: The two discounts are **both on the marked price**, so they multiply ($0.90 \times 0.96 = 0.864$); adding them to "14%" would give a wrong 7.5%.
4. **22.5 L.** $40 \times (1 - 10/40)^2 = 40 \times (3/4)^2 = 40 \times 9/16 = 22.5$. (*Verified.*) The formula holds because the **same** 10 L is removed each round.
5. **66.** Total = $50 \times 60 = 3000$; boys = $30 \times 56 = 1680$; girls' total = 1320 over 20 girls = 66. (*Verified.*) ANALYSIS: The plain mean of 56 and 60 is meaningless here - the groups differ in size.
6. **1000.** $0.8C + 300 = 1.1C \rightarrow 0.3C = 300 \rightarrow C = 1000$. *(*Verified: SP at 20% loss = 800; $800 + 300 = 1100 = 1.1 \times 1000$.*)*

9. PYQ-pattern notes (2024-2026, Set A)

- ANALYSIS: Pure commercial arithmetic is a **lighter** slice of CSAT than candidates fear (a small share of each paper), but it is **high-yield** because the methods are fast and the traps are predictable.
- ANALYSIS: Recurring shapes: **successive percentage change**, **partnership/profit sharing**, **mixtures/alligation** (including replacement), and **ages**.
- ANALYSIS: The paper rewards **multiplier fluency**; the classic mistake it punishes is adding percentages.

10. Study links

- FACT: Foundation companion.
- FACT: Rates, Motion, Time and Geometry - averages/ratios extend to average speed and work rates.
- FACT: Algebra, Inequalities and Data Sufficiency - ages/mixtures are equation setups.

Ownership note: conditional-probability entries in an appended historical routing ledger are owned by Topic 06, not by this Arithmetic companion. This file may supply percentage/fraction prerequisites but must not be used as a probability substitute.

Historical PYQ Integration (2018-2023)

Status: Question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: _PYQ-ROUTING-CSAT-2018-2023.md. **Answer-key rule:** The official 2018-2023 Prelims/CSAT keys are not held locally; no option or answer has been inferred.

- **Years represented:** 2018, 2019, 2020, 2021, 2022, 2023

- **Paper(s):** CSAT

- **Routed question demands:** 40

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2018	CSAT	39	Conditional probability	Objective question; official key unavailable locally	Cross-owner reference; key unavailable locally	Route to Topic 06 for method and timed practice; no Topic 03 coverage claim is made here.
2018	CSAT	41	Pass-fail percentage marks	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	57	Symmetric profit-loss point	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	68	Capacity proportion share	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	69	Compound interest installment	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	11	Percentage weight reduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	17	Percentage shortfall amount	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	19	Unit price joint purchase	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	34	Percentage marks difference	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	39	Discount from quantity change	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	40	Mean error correction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2019	CSAT	55	Average group weight	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	18	Weighted group average	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	39	Age group average	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	40	Loss-percentage cost recovery	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	47	Successive-discount comparison	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	55	Percentage ratio constraint	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	57	Ratio share difference	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	59	Class average invariance	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	68	Successive mixture replacement	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	69	Innings average difference	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	70	Price-hike quantity reduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	78	Volume-mass unit conversion	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	7	Linear fuel-price crossover	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	9	Percentage group overlap	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2021	CSAT	30	Proportional marks threshold	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	37	Class average transfer effect	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	53	Equal-cost price ratio	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	54	Percentage marks comparison	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	55	Conditional money allocation	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	70	Ratio-share inequality	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	78	Successive percentage change	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	8	Denomination combination validity	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	20	Successive percentage change	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	26	Equal interval spacing	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	45	Word problem total cost	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	64	Election vote percentage	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	69	Mixture proportion comparison	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	80	Average weight combination	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2023	CSAT	44	Compound interest frequency comparison	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

What this owner must now support

- Conditional probability - routed to Topic 06 (cross-owner reference)
- Pass-fail percentage marks
- Symmetric profit-loss point
- Capacity proportion share
- Compound interest installment
- Percentage weight reduction
- Percentage shortfall amount
- Unit price joint purchase
- Percentage marks difference
- Discount from quantity change
- Mean error correction
- Average group weight
- Weighted group average
- Age group average
- Loss-percentage cost recovery
- Successive-discount comparison
- Percentage ratio constraint
- Ratio share difference
- Class average invariance
- Successive mixture replacement
- Innings average difference
- Price-hike quantity reduction
- Volume-mass unit conversion
- Linear fuel-price crossover
- Percentage group overlap
- Proportional marks threshold
- Class average transfer effect
- Equal-cost price ratio
- Percentage marks comparison
- Conditional money allocation
- Ratio-share inequality
- Successive percentage change
- Denomination combination validity
- Equal interval spacing
- Word problem total cost
- Election vote percentage
- Mixture proportion comparison
- Average weight combination

- Compound interest frequency comparison

The table integrates the examinable demand and paper metadata. It does not turn an unkeyed objective question into a solved answer, and it does not claim that lexical presence alone proves full conceptual sufficiency.

CONSOLIDATED REGISTER NOTES

Complete revision spine

- **Ratio language:** Convert comparisons into common units and scalable parts.
- **Proportion and variation:** Distinguish direct from inverse change.
- **Percent foundation:** Percent means per hundred; identify the base before operating.
- **Successive change:** Multiply factors; do not add percentages blindly.
- **Average:** Use total divided by count and preserve group weights.
- **Profit, loss and discount:** Keep CP, SP and MP bases distinct.
- **Interest:** Derive SI from principal-rate-time and CI from repeated growth.
- **Mixtures:** Use conservation of quantity or alligation with validity checks.
- **Partnership:** Profit share follows capital multiplied by time.
- **Ages:** Translate one timeline consistently.
- **Estimation and options:** Use smart numbers only when ratios remain invariant.
- **Verification:** Reverse the change, recompute totals and inspect the percentage base.

Ownership and close-option firewall

- **Own here:** Owns ratio, proportion, variation, percentages, averages, mixtures, profit-loss-discount, simple and compound interest, partnership and ages.
- **Do not duplicate:** Pure number properties belong to Topic 02; time-work and motion rates to Topic 04; equation sufficiency to Topic 05.
- **Verification:** Every formula is derived from part/whole, multiplier, weighted-total or principal-time logic. Generated answers are recomputed with exact fractions before formatting.

Timed answer route

CLASSIFY -> EXTRACT -> REPRESENT -> EXECUTE -> VERIFY -> DECIDE

- Use estimation or option elimination only after preserving the governing condition.
- A blank costs zero; a rushed unsupported answer also consumes time and may attract negative marks.
- For every error, record concept/application/calculation/reading/passage/time/guess, repair the owner, and retry a new item rather than memorising the old option.